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*      STAAD.Pro
*      Version 2007   Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=   MAR 1, 2013
*      Time=   16:52: 0
*
*      USER ID: Parsons Binckerhoff
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1. STAAD SPACE
INPUT FILE: 130222 VM1 STAIRCASE 2A.STD
2. START JOB INFORMATION
3. ENGINEER DATE 26-NOV-12
4. JOB NAME VM1 STAIRCASE
5. END JOB INFORMATION
6. INPUT WIDTH 79
7. UNIT METER KN
8. JOINT COORDINATES
9. 1 -109.835 0 -23.25; 2 -109.835 0 -22.05; 3 -107.085 3 -23.25
10. 4 -107.085 3 -22.05; 5 -106.085 0 -23.25; 6 -106.085 0 -22.05
11. 7 -106.085 2 -23.25; 8 -106.085 2 -22.05; 9 -106.085 3 -23.25
12. 10 -106.085 3 -22.05; 11 -101.835 0 -23.25; 12 -101.835 0 -22.05
13. 13 -101.835 3 -23.25; 14 -101.835 3 -22.05; 15 -101.835 4 -23.25
14. 16 -101.835 4 -22.05; 17 -101.835 5 -23.25; 18 -101.835 5 -22.05
15. 21 -102.835 5 -23.25; 22 -102.835 5 -22.05
16. MEMBER INCIDENCES
17. 1 3 1; 2 4 2; 3 4 3; 4 3 7; 5 4 8; 6 9 3; 7 10 4; 8 7 5; 9 8 6; 10 9 6
18. 11 10 5; 12 9 7; 13 10 8; 14 10 9; 15 21 9; 16 22 10; 17 13 11; 18 14 11
19. 19 13 12; 20 14 12; 21 14 13; 22 15 13; 23 16 14; 24 17 14; 25 18 13; 26 17 15
20. 27 18 16; 28 18 17; 34 21 15; 35 22 16; 36 17 21; 37 18 22; 38 22 21
21. DEFINE MATERIAL START
22. ISOTROPIC STEEL
23. E 2.05E+008
24. POISSON 0.3
25. DENSITY 76.8195
26. ALPHA 1.2E-005
27. DAMP 0.03
28. END DEFINE MATERIAL
29. MEMBER PROPERTY JAPANESE
30. 17 20 22 23 26 27 TABLE ST H250X250X9
31. 8 9 12 13 TABLE ST H250X250X9
32. 36 37 TABLE ST H250X250X9
33. 1 TO 7 15 16 TABLE ST C250X90X9
34. 10 11 18 19 24 25 TABLE ST C150X75X6.5
35. 14 21 28 34 35 38 TABLE ST H200X200X8
36. CONSTANTS
37. MATERIAL STEEL ALL
38. BETA 180 MEMB 2 16 TO 28 34 TO 38
39. SUPPORTS
40. 1 2 PINNED
41. 5 6 11 12 FIXED
42. MEMBER RELEASE
43. 15 16 END MY MZ
44. 15 16 START MY MZ
45. 28 38 START MY MZ
46. 21 START MY MZ
47. 21 END MY MZ
48. 28 38 END MY MZ
49. 14 START MY MZ
50. 14 END MY MZ
51. 3 START MY MZ
52. 3 END MY MZ
53. MEMBER TENSION

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54. 18 19
55. MEMBER TENSION
56. 18 19
57. MEMBER TENSION
58. 10 11
59. MEMBER TENSION
60. 24 25
61. LOAD 1 LOADTYPE NONE TITLE DL
62. SELFWEIGHT Y -1 LIST 1 TO 28 34 TO 38
63. MEMBER LOAD
64. *0.75KPA DEAD LOAD*
65. 36 37 UNI GY -0.25
66. 1 2 6 7 15 16 UNI GY -0.25
67. LOAD 2 LOADTYPE NONE TITLE LL
68. *5KPA LIVE LOAD*
69. MEMBER LOAD
70. 36 37 UNI GY -2.5
71. 1 2 6 7 15 16 UNI GY -2.5
72. LOAD 3 LOADTYPE NONE TITLE WLX
73. MEMBER LOAD
74. 10 11 18 19 UNI GX -0.15
75. 1 TO 3 8 9 12 TO 17 20 22 23 26 TO 28 38 UNI GX -0.24
76. LOAD 4 LOADTYPE NONE TITLE WLZ
77. MEMBER LOAD
78. 1 2 6 TO 9 12 13 15 TO 17 20 22 23 26 27 34 TO 37 UNI GZ -0.24
79. LOAD 5 LOADTYPE NONE TITLE NLX
80. SELFWEIGHT X -0.015 LIST 1 TO 28 34 TO 38
81. MEMBER LOAD
82. 36 37 UNI GX -0.004
83. 1 2 6 7 15 16 UNI GX -0.004
84. LOAD 6 LOADTYPE NONE TITLE NLZ
85. SELFWEIGHT Z -0.015 LIST 1 TO 28 34 TO 38
86. MEMBER LOAD
87. 36 37 UNI GZ -0.004
88. 1 2 6 7 15 16 UNI GZ -0.004
89. LOAD 7 LOADTYPE NONE TITLE 1.0DL+1.0LL
90. REPEAT LOAD
91. 1 1.0 2 1.0
92. LOAD 8 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0WLX
93. REPEAT LOAD
94. 1 1.0 2 1.0 3 1.0
95. LOAD 9 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0WLZ
96. REPEAT LOAD
97. 1 1.0 2 1.0 4 1.0
98. LOAD 10 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0NLX
99. REPEAT LOAD
100. 1 1.0 2 1.0 5 1.0
101. LOAD 11 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0NLZ
102. REPEAT LOAD
103. 1 1.0 2 1.0 6 1.0
104. LOAD 21 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 WLX
105. REPEAT LOAD
106. 1 1.0 2 1.0 3 -1.0
107. LOAD 22 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 WLZ
108. REPEAT LOAD
109. 4 -1.0 1 1.0 2 1.0
110. LOAD 23 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 NLX
111. REPEAT LOAD
112. 1 1.0 2 1.0 5 -1.0
113. LOAD 24 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 NLZ
114. REPEAT LOAD
115. 6 -1.0 1 1.0 2 1.0
116. *ULTIMATE LOAD COMBINATIONS
117. LOAD 25 LOADTYPE NONE TITLE 1.0DL+1.6LL
118. REPEAT LOAD
119. 1 1.0 2 1.6
120. LOAD 26 LOADTYPE NONE TITLE 1.4DL+1.6LL
121. REPEAT LOAD
122. 1 1.4 2 1.6
123. LOAD 27 LOADTYPE NONE TITLE 1.2DL+1.2LL+1.2WLX

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124. REPEAT LOAD
125. 1 1.2 2 1.2 3 1.2
126. LOAD 28 LOADTYPE NONE TITLE 1.2DL+1.2LL+1.2WLZ
127. REPEAT LOAD
128. 1 1.2 2 1.2 4 1.2
129. LOAD 29 LOADTYPE NONE TITLE 1.2DL+1.2LL+1.2NLX
130. REPEAT LOAD
131. 1 1.2 2 1.2 5 1.2
132. LOAD 30 LOADTYPE NONE TITLE 1.2LL+1.2LL+1.2NLZ
133. REPEAT LOAD
134. 1 1.2 2 1.2 6 1.2
135. LOAD 41 LOADTYPE NONE TITLE 1.2DL+1.2LL-1.2 WLX
136. REPEAT LOAD
137. 1 1.2 2 1.2 3 -1.2
138. LOAD 42 LOADTYPE NONE TITLE 1.2DL+1.2LL-1.2WLZ
139. REPEAT LOAD
140. 4 -1.2 1 1.2 2 1.2
141. LOAD 43 LOADTYPE NONE TITLE 1.2DL+1.2LL-1.2 NLX
142. REPEAT LOAD
143. 1 1.2 2 1.2 5 -1.2
144. LOAD 44 LOADTYPE NONE TITLE 1.2DL+1.2LL-1.2 NLZ
145. REPEAT LOAD
146. 6 -1.2 1 1.2 2 1.2
147. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 20/ 33/ 6

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 10/ 5/ 33 DOF
TOTAL PRIMARY LOAD CASES = 25, TOTAL DEGREES OF FREEDOM = 90
SIZE OF STIFFNESS MATRIX = 3 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.1/ 4099.6 MB

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 1

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 2

**NOTE-Tension/Compression converged after 1 iterations, Case= 3

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 4

**NOTE-Tension/Compression converged after 1 iterations, Case= 5

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 6

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 7

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 8

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 9

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 10

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 11

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 21

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 22

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 23

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 24

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 25

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 26